Supplementary Material:
Historical Path-Residual Guided Candidate Re-ranking for Catalog-Assisted S-Phase Repicking

1 Algorithm 1: Historical path-residual candidate re-ranking

Algorithm 1 Historical path-residual candidate re-ranking (S phase)

Require: waveform w (current), catalog origin/location (event), station metadata, frozen train-only history store H, frozen PhaseNet STEAD weights, frozen DistanceMLP, $\lambda = (0.5, 2.0, 0.0)$, K=5, k=50

$p_S \leftarrow$ PhaseNet.`annotate`(w) remapped to waveform UTC grid
$C \leftarrow$ top-K peaks of p_S (min distance 50, prominence 0.05, height 0.1; else global argmax)
$\tau^{\text{base}} \leftarrow$ DistanceMLP(distance, depth, ...)
$(n, r_{\text{med}}, r_{\text{mad}}) \leftarrow$ query H with directed path key (train events only)
if $n < 5$ **or** residual-P MAD > 1.0 **then**
    history_available $\leftarrow$ false; $\lambda_h \leftarrow 0$
else
    $\tilde{r} \leftarrow \text{shrink}(r_{\text{med}}, r_{\text{global}}, n, k{=}50)$
    history_available $\leftarrow$ true
end if
$\hat{s}, \sigma \leftarrow$ convert $\tau^{\text{base}} + \tilde{r}$ to sample index / σ_{samp}
for candidate $j \in C$ **do**
    $h_j \leftarrow \exp(-\frac{1}{2}((s_j - \hat{s})/\sigma)^2)$
    $\text{score}_j \leftarrow \lambda_w \log(\pi_j + \varepsilon) + \lambda_h \log(h_j + \varepsilon) + \lambda_\rho \,\text{norm}(\rho_j)$
end for
return $\arg\max_j \text{score}_j$

Input provenance. Current waveform → PhaseNet probabilities and peaks; event catalog → origin time/location for $\hat{\tau}$ and path key; station metadata → path key / distance features; frozen training history → residual median/MAD only (no val/test leakage).

2 Stage 2 development results

Stage 2 mixed fixed-eval (val+test) was used for λ grids and shrinkage search only. See project `artifacts/results/stage2/` and `paper/tables/table_subset_stage2_dev.csv`. Do not treat Stage 2 as the paper primary test.

3 Stage 4 auxiliary table

Regenerate with `scripts/build_paper_assets.py`; CSV mirror: `paper/tables/table_stage4_auxiliary.c`
Underpowered (9 events); bootstrap CIs include zero.

4 Cross-weight transfer

`paper/tables/table_transfer_weights.csv` (STEAD/ETHZ/SCEDC on Stage 4).

5 Learned gate training details (ablation)

Three seeds $\{42, 123, 2026\}$, fewer than 7k parameters; mean gate ≈ 0.97 on Stage 3 test (near-PhaseNet collapse). C-class training examples rare. Not recommended as main method.

6 Oracle analysis

Oracle nearest candidate in $K{=}5$: F1@0.5=0.890 on Stage 3. Oracle PN/fixed selector: 0.720. Gap shows coverage vs selector headroom.

7 Noise / fallback

History-unavailable scoring sets $\lambda_h{=}0$ and matches PhaseNet probability ranking among candidates (identity fallback).

8 Frozen hyperparameters

See `paper/tables/hyperparameters_frozen.json`: $K{=}5$, $k{=}50$, S $\lambda = (0.5, 2.0, 0.0)$, PhaseNet STEAD, pick threshold 0.3, min history 5, MAD disable 1.0 s.

9 Additional figures

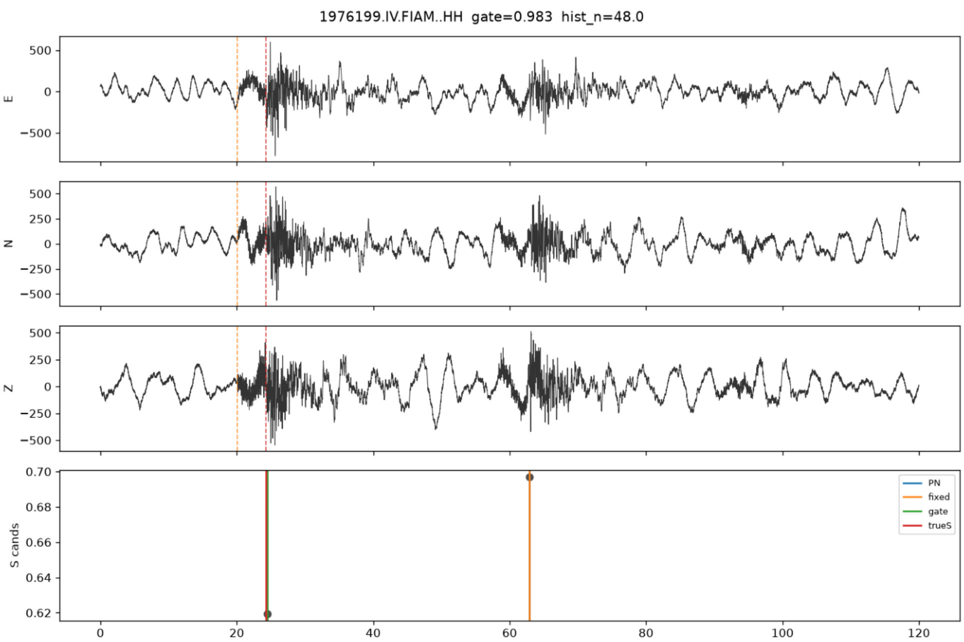

Figure 1: Negative-transfer / fixed-wrong case dump (Stage 3 frozen).

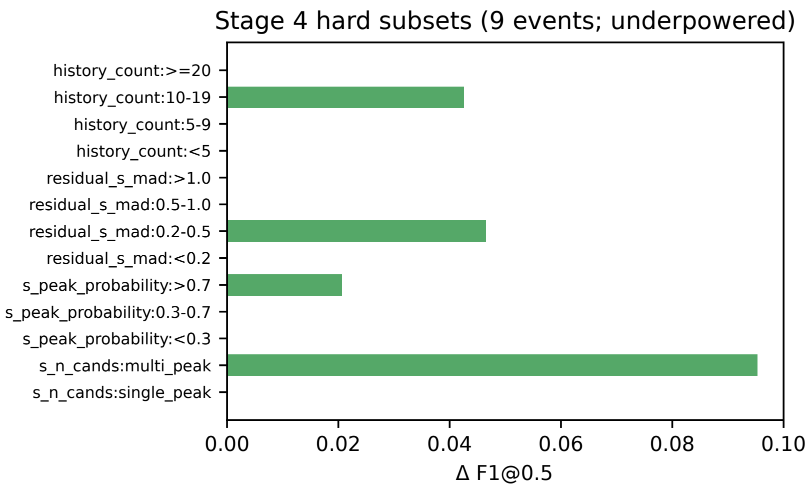

Figure 2: Stage 4 hard-subset deltas (underpowered).